

Software Requirements Specification

Camfranglais

Public lexical and syntactic analysis workspace

DETERMINISTIC / PUBLIC ACCESS / EVIDENCE-LED

Baseline: the implemented five-screen application at source revision `a1b88aa`. Visitors can analyze and save immutable tests without accounts. Collection, saved grammar and recordings are read-only. This specification separates software acceptance from unconfirmed fieldwork provenance.

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1 Purpose and Document Control

This SRS defines the externally observable behavior, constraints and acceptance criteria of Camfranglais for the CS4110 Compiler Construction assignment, Summer 2026, SET A. It is an implemented-product baseline, not a promise of a general-purpose language understanding system. Requirements use “shall” for the current product contract; unimplemented proposals are labelled explicitly.

The companion Software Design Description (SDD) explains the implementation and contains source-grounded UML views. The UML atlas provides separately zoomable diagram sheets. The final coursework report is a distinct 25–30-page document containing the supplied corpus, calculations, test outcomes and genuine browser captures. Its page limit does not apply to this engineering specification or the SDD.

Baseline	Version 4.1; 27 September 2026; public-access release.
Replaces	Version 3.0’s account/private-project, editable-grammar, import and in-app report workflows. Historical archives remain preserved.
Primary stakeholders	Three student authors, instructor/examiner, public visitors, and the deployment operator.
Decision	No login. Everyone may analyze and retain tests; reference Collection, grammar and recordings are not publicly editable.
Change authority	The group and instructor approve requirement changes; the operator controls deployment and offline reference maintenance.

2 Problem, Scope and Evidence

2.1 Educational problem

Informal urban expressions can combine French, English, Pidgin, local slang, nonstandard spellings and incomplete clauses. A conventional compiler’s lexer and parser provide a useful, inspectable model of selected structures, but cannot establish a speaker’s intended meaning, social context or authenticity. Camfranglais therefore preserves raw input and exposes deterministic token classification, grammar transformations, sets, table entries and parse traces.

2.2 Current product boundary

The active navigation contains exactly five destinations:

1. **Franc Analyzer:** type or paste a statement, then analyze and publicly retain its original input and results.
2. **Analysis:** inspect immutable saved tests, vocabulary statistics, separate CFG statistics, saved grammar and computed parser artifacts.
3. **Collection:** search and inspect existing entries, glosses, annotations and available audio without modifying them.
4. **Dictionary:** search source-labelled reference vocabulary.
5. **Synthetic examples:** browse explicitly constructed practice material, never automatically counted as fieldwork.

Login, registration, recovery, Google identity, profiles, logout, project management, public imports, microphone recording, grammar editing, administration, history screens and in-app re-

port/presentation generation are outside the active interface. Translation/chat, OCR, automatic transcription, browser dictation, speech synthesis and AI explanations are not features of this release. Recorded read-aloud uses existing audio; it is not generated speech.

2.3 Corpus and reference baseline

The documentation snapshot contains 13 Collection entries: 12 supplied sentences and one Word annotation. The supplied sentences contain 81 tokens. The saved corpus-specific CFG accepts ten and rejects two, both at token six: `n'ais` and `alli` remain UNKNOWN. These results do not correct the original wording. The reference dictionary exposes 938 entries derived from 878 supplied CSV rows plus retained distinct legacy senses. Twenty-six synthetic examples remain a separate practice library.

These are dated observations, not hard-coded deployment limits. Public test history grows independently of Collection: 34 saved tests were observed online on 27 September. Repeated tests and older snapshots can produce totals that differ from a one-pass evaluation of the twelve statements.

Fieldwork is not certified

The saved manual-transcription confirmation is false and the collection-method field is blank. Some records contain location text or audio references; neither proves when, where or by whom the original expression was heard. The documentation must not invent collection dates, speaker identities, consent or authenticity.

3 Actors, Permissions and Operating Modes

Actor	Permitted actions	Restrictions
Public visitor	View all public pages; analyze input; retain and inspect immutable tests; play available recordings.	No accounts, private user workspace, reference editing or public administration.
Student authors / examiner	The same public actions; use exported documentation and source locally for coursework review.	No elevated browser role is inferred from a name or matricule.
Deployment operator	Configure service, TLS, storage, backups and controlled offline reference maintenance using filesystem/server access.	Not an account role in the public application; maintenance endpoints are not made public.

The default factory mode, `require_auth=None`, selects the public read-only boundary. Explicit `True` retains the authenticated legacy API for compatibility tooling/tests; it is not the deployed public API. Explicit `False` is unrestricted local maintenance and is prohibited in production. Historical classes and records remain in the repository without implying that their interfaces are enabled.

Anonymous cookie and CSRF bootstrap is not authentication. It supplies same-origin request protection, not user identity or access isolation. Public test text is visible to other visitors; the UI shall warn against submitting personal, confidential or sensitive material.

4 Functional Requirements

ID / area	Required behavior	Acceptance criterion
FR-01 Public entry	The application shall open directly into the five-screen workspace without sign-in, registration, SMTP or an identity provider.	A fresh browser loads the analyzer and reference pages without an account. There is no account gate or credential control.
FR-02 Capabilities	Bootstrap shall report public access and allowed capabilities. Reference mutations shall be denied by the server, not merely hidden.	Analyze/save are true; Collection/grammar/recording edit flags are false. Reference mutations return 403; absent maintenance APIs return 404.
FR-03 Input fidelity	The analyzer shall preserve original case, spacing, accents, apostrophes, punctuation and mistakes in the saved input.	Normalization is limited to matching/statistics. Raw saved text is unchanged; tokens retain original spelling. Empty input is a valid test request.
FR-04 Lexer	The lexer shall segment with regular expressions, then classify deterministically using structural rules, approved Word annotations, explicit word lists, reference categories, French forms and morphological fallbacks.	Unsupported non-whitespace input remains represented as UNKNOWN. Conflicting reference classifications produce AMBIGUOUS, not guessed context.
FR-05 Language clues	Code-mixing clues shall compare French/English/Pidgin language-candidate sets independently of token part-of-speech labels.	Unknowns carry no language evidence. Sentence-ending punctuation resets the comparison. Output retains the original endpoint spellings.
FR-06 Phrase annotations	Recognized multiword verb/slang expressions shall be additional source-ordered annotations, not replacements for the original token stream.	Phrase boundaries exclude word-internal matches; absence of a phrase match does not erase, merge or reject otherwise valid tokens.
FR-07 Vocabulary verdict	Vocabulary approval shall reject exactly tests with one or more UNKNOWN tokens. It shall remain separate from CFG acceptance.	Recognized slang, AMBIGUOUS and other known labels pass vocabulary approval. Empty input has no UNKNOWN, but the saved corpus CFG still rejects it.
FR-08 Saved grammar	New public analyses shall use the operator-saved grammar, displayed read-only.	A stale/different grammar on a new analysis returns 409 with a reload instruction; the browser cannot submit an editable local draft as the shared grammar.

ID / area	Required behavior	Acceptance criterion
FR-09 CFG transforms	The grammar service shall parse bounded notation, perform supported direct and ordered indirect left-recursion elimination, then left-factor.	Each transformation records before/after rules and its operation. Unsafe cycles, notation errors and excessive expansion fail explicitly.
FR-10 Sets and table	FIRST/FOLLOW shall be computed to a fixed point, including epsilon and end-of-input. The LL(1) table shall retain conflict diagnostics.	No conflicting production is silently selected. A residual recursive or conflicting grammar disables predictive parsing.
FR-11 Predictive parser	A bounded stack machine shall consume lexical categories using the LL(1) table and expose stack, remaining input, action and consumed-token count.	The rightmost stack symbol is the top; snapshots precede the action. Acceptance requires complete input consumption. Failure identifies the position.
FR-12 Immutable tests	Analyze shall retain the input, grammar, lexical results, statistics, approval and parser trace as one original test snapshot.	Refresh preserves tests. No public edit/delete route exists. Later dictionary/grammar changes do not rewrite earlier results.
FR-13 Retry semantics	A request UUID shall distinguish retrying a submission from intentionally running a new test.	An identical retry returns the original record, including after an operator grammar change. Conflicting reuse fails; a new completed test counts separately.
FR-14 Statistics	Analysis shall aggregate original saved-test snapshots, including repeats; Collection entries not tested shall not inflate these statistics.	Vocabulary and CFG totals are visibly distinct. Frequency case-folding and orthographic variation rules are described; unknown counts remain explicit.
FR-15 Collection	Visitors shall search/filter and view existing raw expressions, glosses, annotations and available attachments read-only.	No create/edit/delete/import control is active. Existing raw text, revisions and ownership are preserved in storage during the public-access upgrade.
FR-16 References	Dictionary entries shall retain source/line, origin, part of speech and distinct senses. Examples shall be clearly synthetic.	Search does not turn dictionary entries/examples into collected statements. Ambiguous senses and reference-origin limitations are visible.
FR-17 Recorded audio	Read-aloud shall look up and play only an existing compatible recording, with an explicit unavailable state otherwise.	No TTS fallback, browser speech synthesis, voice cloning or public upload/ record/delete function is invoked. Playback failures are actionable.

ID / area	Required behavior	Acceptance criterion
FR-18 Ownership privacy	Public responses shall redact historical ownership presentation without altering stored owners. New tests use the internal public actor convention.	Public ownership is null/blank and cannot edit. No fake user account is created. Existing credentials and archival records remain untouched.
FR-19 Recovery boundary	Configured operator backup lifecycle shall continue privately. The release shall preserve existing databases/media and immutable tests.	Backup/restore/admin APIs are not mounted publicly. Operators verify protected, database-consistent backups and rehearse restoration in isolation.
FR-20 Documents	Source-grounded SRS/SDD/UML and a separate 25–30-page coursework report shall be supplied in editable form and PDF.	All production classes have UML coverage; sequence activations balance; current images are embedded; PDF references/layout pass the strict checker.

5 Quality Requirements and Failure Contracts

ID	Requirement and verification boundary
QR-01 Correctness	The same bounded input, grammar and lexicon snapshot shall yield the same compiler output. Tests shall exercise actual source sentences and independent boundary cases. Passing a small corpus is not an accuracy benchmark.
QR-02 Durability	SQLite transactions and coordinated locks shall protect shared snapshots. Persistence failures shall never be presented as successful saves. Browser cancellation does not guarantee that the server cancelled an in-flight save.
QR-03 Public safety	HTTPS, trusted hosts/proxies, body limits, same-origin CSRF, secure production cookies, rate limits and security headers shall remain. These controls do not make public submissions private or constitute an abuse-proof system.
QR-04 Explainability	The UI and report shall distinguish vocabulary recognition, grammar fit, heuristic language evidence, supplied glosses and verified facts. No generated meaning or hidden correction may substitute for a failed analysis.
QR-05 Accessibility	Keep labelled controls, keyboard navigation, visible errors, focus management, responsive layouts and text alternatives to audio. Tests on selected desktop/ mobile browsers do not certify all devices or assistive technologies.
QR-06 Maintain-ability	Use the existing Python/TypeScript test suites, source-labelled references, local PlantUML renderer and LaTeX pipeline. No online renderer receives project code or private records. Historical evidence is preserved with its original date.

ID	Requirement and verification boundary
QR-07	Bound input, grammar expansion, parser traces and retained media.
Operational bounds	No unmeasured latency, throughput, uptime or concurrency guarantee is claimed. Public tests have no automatic purge: operators must monitor storage growth.

Failure	Required distinction
403	Invalid Origin/CSRF or a forbidden reference mutation.
404	Missing resource or an endpoint not exposed in public mode.
409	Stale grammar or conflicting idempotency input, with recovery guidance.
413 / 422	Oversized or structurally invalid input, not successful analysis.
429	Client rate limit exceeded, with Retry-After .
503	Operational/storage/reference failure, explicitly surfaced.
CFG rejection	A successfully computed result, not a network failure; it may be retained as an immutable test.

6 Data and Lifecycle Requirements

6.1 Logical records

Record	Purpose and lifecycle	Public policy
Collection entry	Raw wording, glosses, reviewed annotations, location text and audio references; revisions preserved.	Read-only.
Saved grammar/profile	Grammar, rationale, limitations and retained coursework metadata in shared storage.	Grammar read-only; profile management not exposed.
Analyzer test	Original raw text, grammar/lexer/parser snapshots, derived statistics, timestamp and request identity.	Create and read; immutable; public.
Recorded reading	Existing saved audio with matching text and metadata; separate from Collection audio attachments.	Lookup and playback only.
Dictionary and examples	Versioned bundled references and synthetic examples.	Read-only; not fieldwork.
Legacy accounts/history	Existing credentials, identities, older workspace archives and histories retained for compatibility.	Never served by the default public API or as static assets.
Operational records	Public rate counters, private backups and server configuration.	Operator-only; not application content.

The default local data root is `.mboa`; the shared workspace resides under `shared-workspace/workspace.sqlite3`. Public rate counters use `public-access.sqlite3`, separate from the legacy authentication database. Deployment uses a persistent directory outside the web root. Public response redaction does not delete or replace the original stored ownership.

Backups are sensitive recovery artifacts, not evidence of record authorship. The current shared workspace document format is 5; the outer archive manifest remains format 1. Backward-compatible migration and validation must preserve older formats and related media. Same-disk backups are not disaster recovery. Credentials/configuration require separately protected full-root backup.

6.2 Active processing bounds

Boundary	Current limit
General JSON body	1 MiB.
Statement / grammar text	4,000 / 12,000 characters.
Coursework corpus helper	500 entries / 100,000 text characters.
Parser input / token text	256 tokens / 1,000 characters per token.
Parse trace / stack	2,000 steps / 1,024 symbols.
Transformed grammar	200 nonterminals; 600 alternatives; 6,000 RHS symbols.
Anonymous request rate	120 requests per 60 seconds per client address; health excluded.

The corpus-helper bounds are not the assignment target of 10–15 statements, nor a guarantee that every maximum-size grammar will fit transformation budgets. Additional legacy import/media/backup guards remain in their private code paths; they are not public upload features. Visitors sharing an address may share a rate bucket; trusted proxy configuration affects address attribution.

7 External Interfaces

Area	Default public interface
Bootstrap	GET /api/public/session: public capability policy, anonymous CSRF token and cookie; not an account session.
Single lexer / analysis	POST /api/analyze and /api/analyzer/analyze; same-origin CSRF protection.
Saved tests	POST /api/analyzer/tests; read list/report/detail through the analyzer routes. No public test-update/delete interface.
Read-only data	Metadata, dataset/search and available audio, dictionary, examples and saved analyzer state/grammar.
Readings	POST /api/readings/lookup; existing reading byte playback. No public record, upload, replacement or deletion.
Health	GET /api/health: status=ok, mode=compiler; no external language-provider dependency.
Not mounted	/api/auth/*, /api/workspace/*, /api/coursework/*, public imports, restore and administration.

The anonymous cookie is HttpOnly and SameSite=Lax, with Secure in production. Public POST

requests require the allowed Origin and X-CSRF-Token. No project or account selection is needed in the browser. Public resources are not protected by an authentication wall; confidentiality cannot be inferred from CSRF protection.

8 Deployment and Operational Acceptance

The selected deployment is a single durable Ubuntu VPS, supervised by systemd and fronted by Nginx. The public address is <https://camfranglais.duckdns.org>; the application listens only on 127.0.0.1:2021. Port 2021 is an internal upstream, not an extra public-facing URL. HTTPS traffic uses port 443; port 80 supports the configured HTTP redirect/certificate flow.

Application	/var/www/francanglaiscompiler
Persistent data	/var/lib/camfranglais
Environment	/etc/camfranglais.env
Service	camfranglais.service, under a restricted service account
Required production settings	MBOA_ENVIRONMENT=production, HTTPS MBOA_PUBLIC_URL, absolute MBOA_DATA_DIR
Proxy	Preserve Host and forwarded protocol/client address; trust only the actual reverse proxy.

SMTP/Google configuration is not required in public mode. The operator shall preserve existing environment and data, validate Nginx before reload, protect off-host backups and avoid changing unrelated sites on the shared VPS. Interactive API documentation is disabled in production. The documentation session observed the public bootstrap returning HTTP 200 and the direct no-login browser interface on 27 September; this does not prove long-term availability, backup recovery or the mechanism used to deploy it.

9 Verification and Traceability

Requirement group	Evidence / check	Limitation
FR-01–02, 18–19	Public-boundary backend tests; fresh desktop/mobile browser workflows; observed live no-login policy.	No full independent security audit or availability certification claimed.
FR-03–07	Lexer/language/reference tests; exact twelve-sentence token fixtures; saved lexical snapshots.	Reference labels and language clues are not independent linguistic ground truth.
FR-08–11	Grammar/transformation/parser tests; all twelve own-data outcomes; full FIRST/FOLLOW and LL(1) table in the report.	Category grammar has restricted coverage; positive matches can still be nonstandard speech.
FR-12–14	Idempotency, immutability, persisted history/statistics tests and isolated data-preservation checks.	Growing public history differs from a fixed controlled corpus.

Requirement group	Evidence / check	Limitation
FR-15–17	Read-only Collection/grammar/audio tests; live reference views and recorded playback regressions.	A recording is not proof of original fieldwork.
FR-20	Local PlantUML/LaTeX build; strict class, activation, image and PDF checks; measured report pagination.	Human authors still review provenance, attribution and submission requirements.

The 26 September release record reports 186 backend/launcher/legacy-storage tests, 148 frontend unit tests, 68 mocked browser tests, passing TypeScript checks and a production build. Fourteen distinct live workflows passed across the full run and focused rerun: two startup/teardown interruptions occurred in the full run and passed in a three-case mobile rerun. This is not an uninterrupted 14/14 result.

The seven targeted corpus grammar/persistence tests passed again on 27 September. One known pre-existing full-suite regression remains: `compiler.tests.test_parser.GrammarNotationTests.test_every_real_lexer_category_is_a_supported_terminal`. Its old expected category set omits newly supported categories. This documentation task does not repair it and does not claim the whole CI suite is green.

10 Submission Gates and Exclusions

1. The group shall confirm the true origin and manual-transcription process, or clearly retain the provenance limitation in the submitted report.
2. The authors shall review original wording, supplied glosses, lexical ambiguities, the two unresolved tokens and the corpus-specific CFG rationale.
3. The final report shall contain the raw statements, actual token tables, regex rules, transformations, FIRST/FOLLOW, the complete LL(1) table, parser results, real screenshots and linguistic discussion.
4. The built report shall measure 25–30 PDF pages, including front matter. The SRS, SDD and UML atlas are supplementary engineering documents and are not included in that page limit.
5. The required ten-minute presentation/demo and individual contributions shall be rehearsed separately. This documentation release does not claim that slides, oral performance, authenticity or instructor approval are complete.

Meaning of acceptance

Software verification establishes behavior on stated tests and snapshots. It cannot certify speaker intent, accurate transcription of an unheard event, fieldwork consent, population-level language coverage or academic approval.